

Mitchell Dickau, PhD

Climate Scientist |
Scientific Programmer |
Earth System Modeler |
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Climate scientist with 7+ years of programming and research experience specializing in Earth system modeling, climate adaptation, and Earth system response to positive and negative emissions. Proven track record of automating scientific workflows, analyzing large datasets, and building interactive data products in Python. Lead author in top-tier journals and a contributor to Canada’s national climate assessment. Looking to leverage analysis and scientific research to drive real-world climate solutions.

WORK EXPERIENCE

CLIMATE ADAPTATION RESEARCHER	ENVIRONMENT AND CLIMATE CHANGE CANADA (Montréal)	
	CANADIAN CENTRE FOR CLIMATE SERVICES DATA PRODUCTS OFFICE	2023-2025
	• Interactive Dashboard Development: Built a web app using Python (Panel, HoloViz, Xarray) to visualize fire weather projections, serving municipal governments, insurance and reinsurance firms, and fire management stakeholders. • Data Quality & Due Diligence: Evaluated novel fire projection datasets for client readiness, performing rigorous quality control and vetoing flawed data products to safeguard institutional credibility. • Multi-Terabyte Data Pipelines: Engineered automated Python workflows (Xarray, NumPy) to process 10+ TB of CMIP6 and CanDCS-M6 downscaled climate ensembles, calculating customized metrics for client-driven adaptation requests. • Collaboration: Partnered with internal outreach teams and external climate service agencies to translate high-dimensional Earth system data into accessible, decision-ready public resources.	
POSTDOCTORAL FELLOW	CONCORDIA UNIVERSITY (Montréal)	
	2026-PRESENT	
PHD RESEARCHER	• Science Synthesis: Invited to co-author the 10 New Insights in Climate Science study led by Future Earth (UN) and the World Climate Research Programme. My lead-author research on temperature overshoot was selected as one of the featured new insights. • Net-Zero Energy Modeling: Evaluated Canadian electrification pathways, conducting quantitative analysis on energy trajectories required to achieve net-zero targets by 2050.	
	2019-2025	
	• HPC & Pipeline Automation: Executed parallel climate model simulations across High-Performance Computing (HPC) clusters using Bash scripts and automated Python pipelines (Xarray, Pandas), processing multi-terabyte Earth system NetCDF datasets. • Carbon & Climate Risk Frameworks: Developed quantitative frameworks (such as degree-years of temperature overshoot and storage-year carbon accounting) to evaluate temporary land carbon storage and inform net-zero policy targets. • Geospatial and Statistical Analysis: Designed custom and scalable workflows in Python to analyze Earth system responses for published peer-reviewed studies. • National Climate Assessment: Co-authored a section for Canada’s Changing Climate Report (slated for Fall 2026 release), detailing emission scenario development.	
	2019-2025	
COURSE INSTRUCTOR & TA	• Instruction & Technical Training: Taught undergraduate physical geography on atmosphere and hydrosphere (125 students), guest lectured graduate-level classes, and taught practical labs in geospatial programming, climate data analysis, and GIS (Python, R, ArcGIS Pro, Terrset).	
	2018-2019	
RESEARCH ASSISTANT	• Technical Project Management: Led a team to complete the end-to-end build of an interactive carbon emissions calculator web app, translating complex climate metrics into technical specifications and simple user-facing tools.	

**CLIMATE
CONSULTANT**

- **Domain Expertise & AI Alignment:** Provided subject-matter expertise on climate science to inform the development of an interactive climate-focused AI chatbot.

TERRAM VERO CONSULTING (Vancouver)

2018-2019

DATA ANALYST

- **Mineral Analysis:** Conducted quantitative analysis on mineral core samples by taking high-resolution images and analyzing the mineral content in mineral cross sections using Python.
- **Automated Data Processing & Analysis:** Automated routine quality control workflows in R to streamline modelling synthesis reports, reducing a 4hr workflow into an executable script running in less than a minute.

SKILLS

- **Public Engagement & Science Communication:** Scientific communication (10+ TV/Radio interviews, Op-Ed), International conference presentations (EGU, AGU, IUGG)
- **Climate Analytics:** Earth System Modeling, scenario analysis, climate risk & adaptation, carbon cycle dynamics
- **Programming:** Python, R, Bash, Git, HPC / Cluster computing
- **Data Science & Geospatial Stack:** Xarray, Pandas, NumPy, SciPy, GeoPandas, Cartopy, Panel, HoloViz, Matplotlib, NetCDF, ArcGIS Pro
- **Languages:** English (Fluent), French (Advanced)

EDUCATION**PhD - Geography, Urban and Environmental Studies | Concordia University (2019-2025)**

- Thesis: Informing climate policy through advancements in our understanding of the Earth system response to emissions, temperature overshoot, and temporary carbon storage
- Graduate Research Communicator of the Year, Concordia University (2024)
- NSERC Doctoral Scholarship (2021-2024), FRQNT Master's Scholarship (2019), Concordia Faculty of Arts and Science Doctoral Scholarship (2020-2024), Hydro-Quebec Master's Scholarship (2019)

BA Honours - Geography (Minor in Sociology) | Concordia University (2013-2018)

- Top 1% undergrad 2017 | Graduated with distinction | Dean's List

PEER-REVIEWED PUBLICATIONS

1. **Dickau, M.** & Matthews, H. D. (Accepted, 2026). Temporary land carbon storage can avoid slow-responding climate changes. *Nature Communications*. [Link](#)
2. Ospina, D., et al. (In revision, 2026). Ten new insights in climate science 2026. *Global Sustainability*.
3. **Dickau, M. et al.**, (2026). Irreversible climate changes driven by degree-years of temperature overshoot. *Communications Earth & Environment*. [Link](#)
4. Wynes, S., ... **Dickau, M.** et al. (2024) Perceptions of carbon dioxide emission reductions and future warming among climate experts. *Communications Earth & Environment*. [Link](#)
5. Wynes, S., **Dickau, M.** et al., (2022). Frequent pro-climate messaging does not predict pro-climate voting by United States legislators. *Environmental Research: Climate*. [Link](#)
6. **Dickau, M.** et al., (2022). The Role of Remaining Carbon Budgets and Net-Zero CO2 Targets in Climate Mitigation Policy. *Current Climate Change Reports*. [Link](#)
7. Matthews, H. D. ... **Dickau, M.** et al., (2022). Temporary nature-based carbon removal can lower peak warming in a well-below 2 °C scenario. *Communications Earth & Environment*. [Link](#)
8. **Dickau, M.** et al., (2020). Projections of declining outdoor skating availability in Montreal due to global warming. *Environmental Research Communications*. [Link](#)

REPORTS

1. Environment and Climate Change Canada. (2026). [Chapter 3: Overview of future changes in climate](#). In Canada's Changing Climate Report 2026. Government of Canada.